



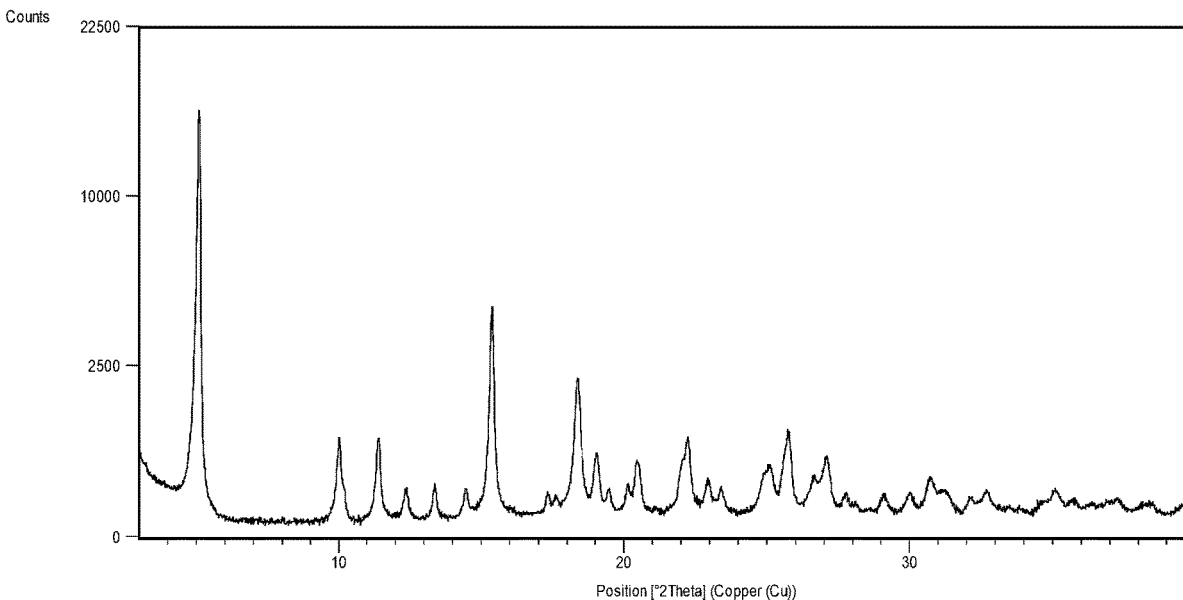
US 20220009893A1

(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2022/0009893 A1**
(43) **Pub. Date: Jan. 13, 2022**(54) **SOLID STATE FORMS OF REPROXALAP**(30) **Foreign Application Priority Data**(71) Applicant: **TEVA PHARMACEUTICALS
INTERNATIONAL GMBH**, Jona
(CH)Dec. 12, 2018 (IN) 201811047061
Feb. 22, 2019 (IN) 201911007092(72) Inventors: **Anantha Rajmohan Muthusamy**,
Sivakasi (IN); **Sundara Lakshmi**
Kanniah, Vellore (IN); **Rahul Kumar**
Reddy Putikum, Mahabubnagar (IN);
Sudhanshu Saxena, Mumbai (IN);
Amol Bandal, Raigad District (IN)**Publication Classification**(51) **Int. Cl.**
C07D 215/38 (2006.01)
(52) **U.S. Cl.**
CPC **C07D 215/38** (2013.01); **C07B 2200/13**
(2013.01)(21) Appl. No.: **17/311,746**(22) PCT Filed: **Dec. 12, 2019**(86) PCT No.: **PCT/US2019/065836**

§ 371 (c)(1),

(2) Date: **Jun. 8, 2021**(57) **ABSTRACT**

The present disclosure relates to solid state forms of Reproxalap salts, processes for the preparation thereof, pharmaceutical formulations/compositions thereof, and methods of use thereof.



An X-ray powder diffraction pattern (XRPD) of Form U1 of Reproxalap 1,
2-ethanedisulfonate

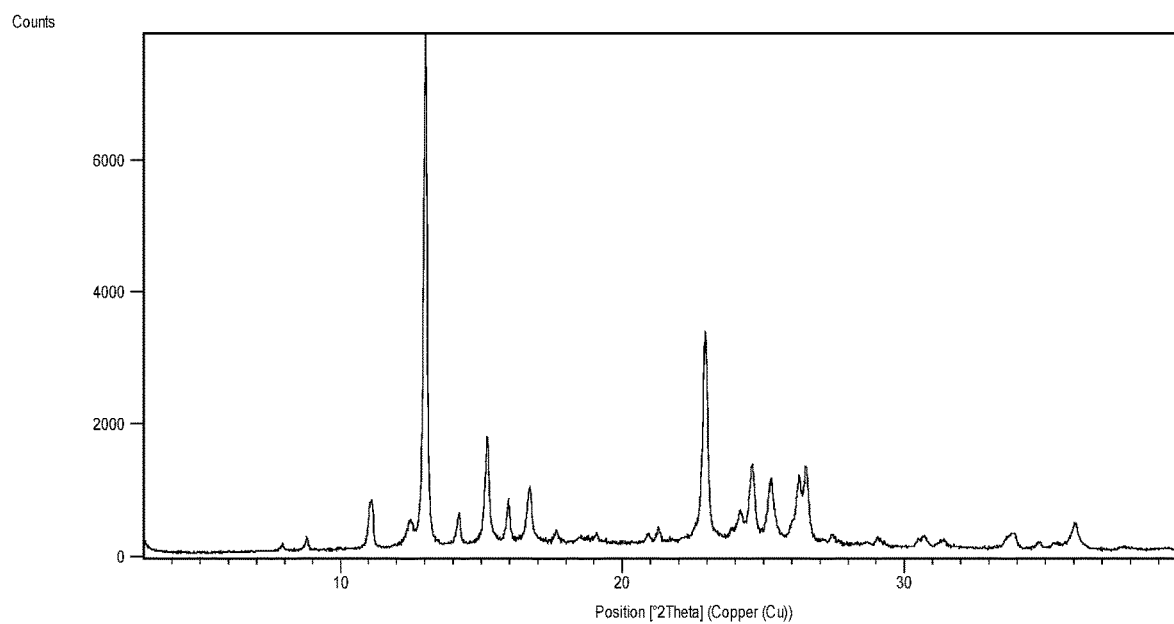


Figure 1: An X-ray powder diffraction pattern (XRPD) of Form R1 of Reproxalap borate.

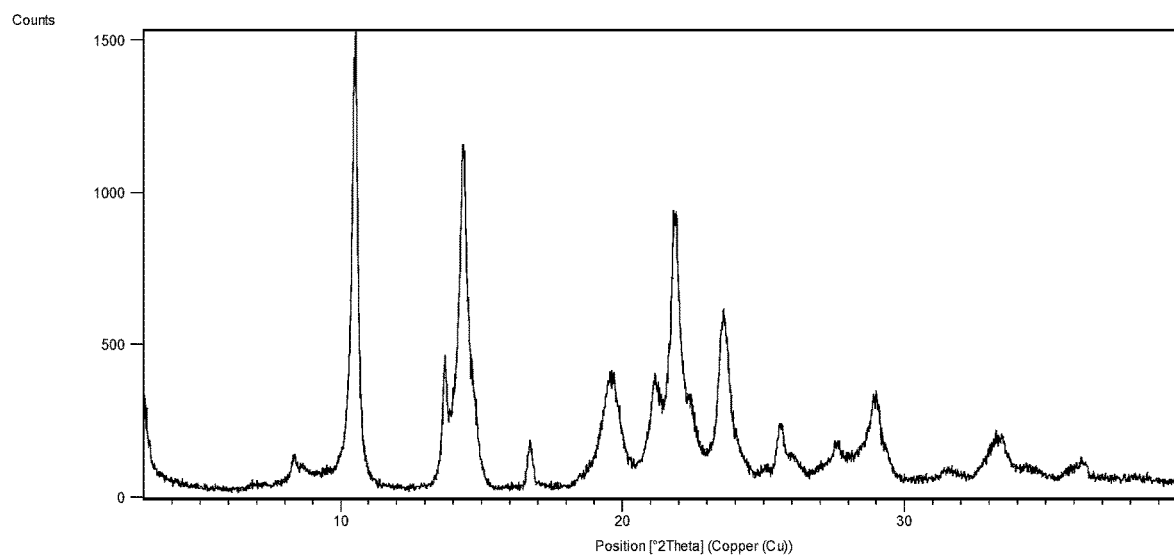


Figure 2: An X-ray powder diffraction pattern (XRPD) of Form S1 of Reproxalap 1,5-naphthalenedisulfonate

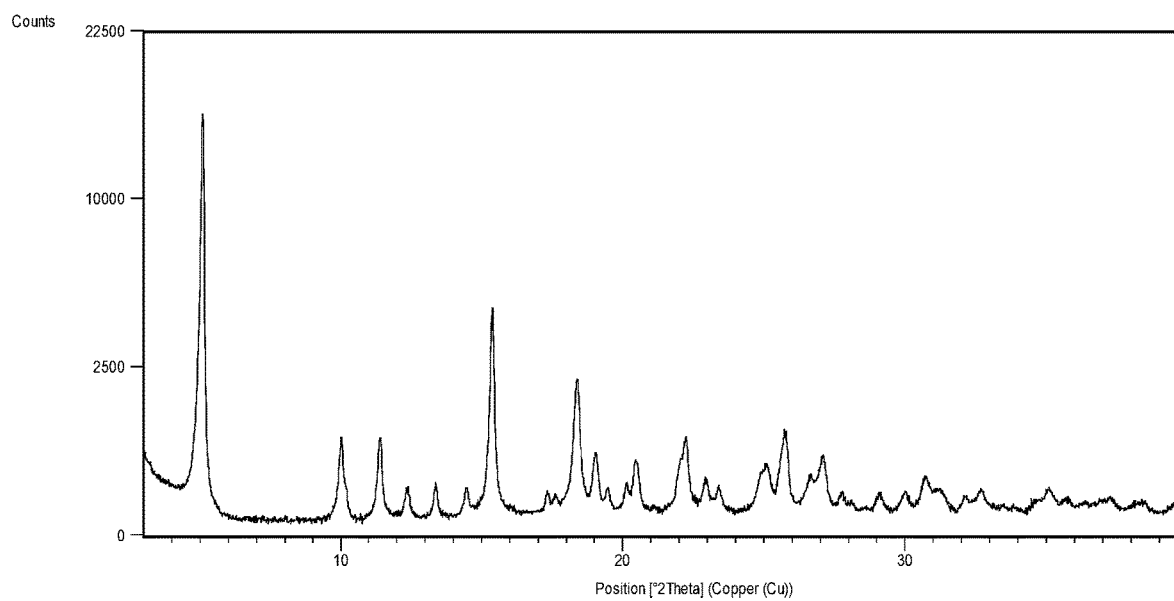


Figure 3: An X-ray powder diffraction pattern (XRPD) of Form U1 of Reproxalap 1, 2-ethanesulfonate

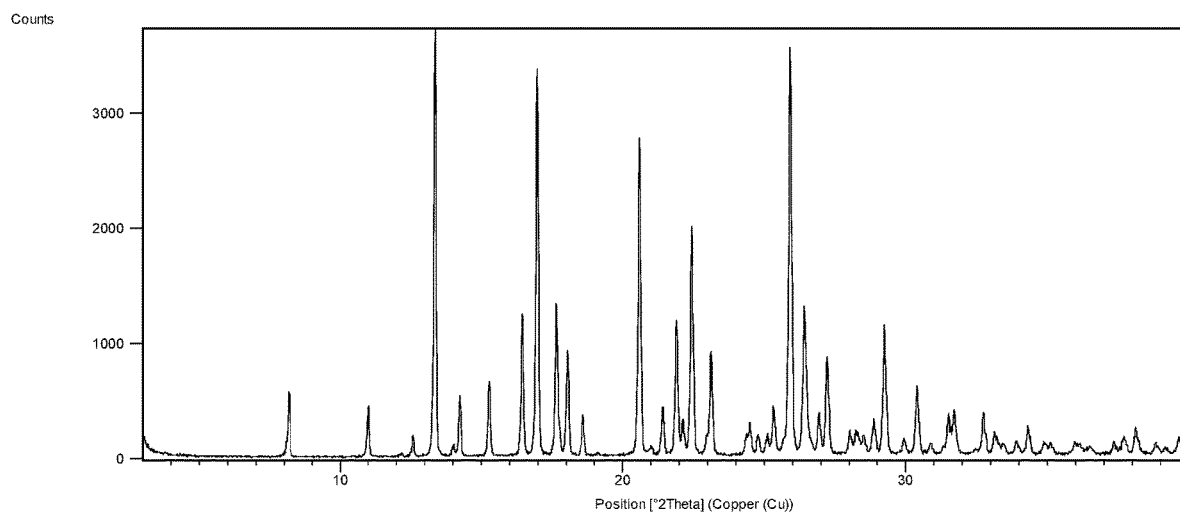


Figure 4: An X-ray powder diffraction pattern (XRPD) of Form P1 of Reproxalap methanesulfonate

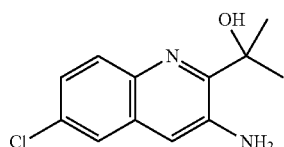
SOLID STATE FORMS OF REPROXALAP

TECHNICAL FIELD

[0001] The present disclosure relates to solid state forms of Reproxalap salts, processes for the preparation thereof, pharmaceutical formulations/compositions thereof, and methods of use thereof.

BACKGROUND

[0002] Reproxalap chemical name is 2-(3-amino-6-chloroquinolin-2-yl)propan-2-ol, having the following chemical structure:



[0003] Reproxalap is a small molecule RASP (Reactive Aldehyde Species) inhibitor under development for the treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis, and Sjögren-Larsson Syndrome.

[0004] U.S. Pat. No. 7,973,025 (referred to herein as '025) discloses Reproxalap. U.S. Patent Application Publication Number 2013/0190500 (referred to herein as '500) discloses a process for preparation of Reproxalap. U.S. Patent Application Publication Number 2018/0265474 (referred to herein as '474) discloses mesylate, besylate, sulfate, tosylate, hydrochloride, oxalate, phosphate, camsylate and tartrate salts and polymorphs of Reproxalap.

[0005] Polymorphism, the occurrence of different crystal forms, is a property of some molecules and molecular complexes. A single compound, like Reproxalap or a salt thereof, may give rise to a variety of polymorphs having distinct crystal structures and physical properties like melting point, thermal behaviors (e.g. measured by thermogravimetric analysis—"TGA", or differential scanning calorimetry—"DSC"), X-ray powder diffraction (XRPD) pattern, infrared absorption fingerprint, Raman absorption fingerprint, and solid state (^{13}C -) NMR spectrum. One or more of these techniques may be used to distinguish different polymorphic forms of a compound.

[0006] Different salts and solid state forms (including solvated forms) of an active pharmaceutical ingredient may possess different properties. Such variations in the properties of different salts and solid state forms and solvates may provide a basis for improving formulation, for example, by facilitating better processing or handling characteristics, improving the dissolution profile, or improving stability (polymorphic as well as chemical stability) and shelf-life. These variations in the properties of different salts and solid state forms may also provide improvements to the final dosage form, for instance, if they serve to improve bioavailability. Different salts and solid state forms and solvates of an active pharmaceutical ingredient may also give rise to a variety of polymorphs or crystalline forms, which may in turn provide additional opportunities to use variations in the properties and characteristics of a solid active pharmaceutical ingredient for providing an improved product.

[0007] Discovering new salts and solid state forms and solvates of a pharmaceutical product can provide materials having desirable processing properties, such as ease of handling, ease of processing, storage stability, and ease of purification or as desirable intermediate crystal forms that facilitate conversion to other salts or polymorphic forms. New polymorphic forms and solvates of a pharmaceutically useful compound can also provide an opportunity to improve the performance characteristics of a pharmaceutical product (dissolution profile, bioavailability, etc.). It enlarges the repertoire of materials that a formulation scientist has available for formulation optimization, for example by providing a product with different properties, e.g., a different crystal habit, higher crystallinity or polymorphic stability which may offer better processing or handling characteristics, improved dissolution profile, or improved shelf-life. For at least these reasons, there is a need for additional solid state forms (including solvated forms) of Reproxalap salts.

SUMMARY OF THE DISCLOSURE

[0008] The present disclosure relates to crystalline forms of Reproxalap borate, Reproxalap 1,5-naphthalenedisulfonate, Reproxalap 1,2-ethanedisulfonate and/or Reproxalap methanesulfonate (may also referred to as Reproxalap mesylate).

[0009] The present disclosure also relates to the uses of solid state forms of Reproxalap salts of the present disclosure, for preparing other salts and their solid state forms, as well as solid state forms of Reproxalap.

[0010] The present disclosure also encompasses the uses of the above described solid state forms of Reproxalap salts for the preparation of pharmaceutical compositions and/or formulations.

[0011] In another embodiment, the present disclosure encompasses pharmaceutical compositions comprising the any one or a combination of the above described solid state forms of Reproxalap salts.

[0012] In yet another embodiment, the present disclosure encompasses pharmaceutical formulations comprising any one or a combination of the above described solid state forms of Reproxalap salts, and at least one pharmaceutically acceptable excipient. The pharmaceutical formulations may be in the form of ophthalmic solution or a topical cream and/or an ointment.

[0013] The present disclosure further encompasses processes to prepare said pharmaceutical formulations of solid state forms of Reproxalap salts comprising combining any one or a combination of the above described solid state forms of Reproxalap salts, or pharmaceutical compositions comprising them, and at least one pharmaceutically acceptable excipient.

[0014] The solid state forms of Reproxalap salts defined herein as well as the pharmaceutical compositions and formulations of the solid state form of the Reproxalap salts can be used as medicaments, particularly for the treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis, and Sjögren-Larsson Syndrome, comprising administering a therapeutically effective amount of any one or a combination of the solid state form of Reproxalap salts of the present disclosure, or at least one of the above pharmaceutical compositions or formulations, to a subject which needs the treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis, and/or Sjögren-Larsson Syndrome, or otherwise in need of the

treatment. The pharmaceutical formulations may be in the form of ophthalmic solution or a topical cream and/or an ointment. In embodiments, the ophthalmic solution is used for the treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis, and/or Sjögren-Larsson Syndrome. In other embodiments, the topical cream and/or an ointment is used for the treatment of Sjögren-Larsson Syndrome.

[0015] The present disclosure also provides the uses of the solid state forms of Reproxalap salts of the present disclosure, or at least one of the above pharmaceutical compositions or formulations, for the manufacture of medicaments for the treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis, and/or Sjögren-Larsson Syndrome.

[0016] In another aspect, the present disclosure encompasses the above solid state forms of Reproxalap salts for use in medicine, preferably for the treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis, and/or Sjögren-Larsson Syndrome.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 shows a characteristic X-ray powder diffraction pattern (XRPD) of Form R1 of Reproxalap borate;

[0018] FIG. 2 shows a characteristic X-ray powder diffraction pattern (XRPD) of Form S1 of Reproxalap 1,5-naphthalenedisulfonate;

[0019] FIG. 3 shows a characteristic X-ray powder diffraction pattern (XRPD) of Form U1 of Reproxalap 1,2-ethanedisulfonate; and

[0020] FIG. 4 shows a characteristic X-ray powder diffraction pattern (XRPD) of Form P1 of Reproxalap methanesulfonate.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0021] The present disclosure relates to solid state forms of Reproxalap salts, to processes for preparation thereof and to pharmaceutical compositions and formulations comprising these solid state forms and/or combinations thereof. Particularly, the present disclosure relates to solid state forms of Reproxalap borate, Reproxalap 1,5-naphthalenedisulfonate, Reproxalap 1,2-ethanedisulfonate and Reproxalap methanesulfonate.

[0022] The solid state forms of Reproxalap salts according to the present disclosure may have advantageous properties selected from at least one of: chemical or polymorphic purity, flowability, solubility, dissolution rate, bioavailability, morphology or crystal habit, stability—such as chemical stability as well as thermal and mechanical stability with respect to polymorphic conversion, stability towards dehydration and/or storage stability, a lower degree of hygroscopicity, low content of residual solvents, adhesive tendencies and advantageous processing and handling characteristics such as compressibility, and bulk density.

[0023] A crystal form may be referred to herein as being characterized by graphical data “as depicted in” a Figure. Such data include, for example, powder X-ray diffractograms and solid state NMR spectra. As is well-known in the art, the graphical data potentially provides additional technical information to further define the respective solid state form (a so-called “fingerprint”) which can not necessarily be described by reference to numerical values or peak positions

alone. In any event, the skilled person will understand that such graphical representations of data may be subject to small variations, e.g., in peak relative intensities and peak positions due to factors such as variations in instrument response and variations in sample concentration and purity, which are well known to the skilled person. Nonetheless, the skilled person would readily be capable of comparing the graphical data in the Figures herein with graphical data generated for an unknown crystal form and confirm whether the two sets of graphical data are characterizing the same crystal form or two different crystal forms.

[0024] A crystal form of Reproxalap salt, referred to herein as being characterized by graphical data “as depicted in” a Figure will thus be understood to include any crystal form of the Reproxalap salt, characterized with the graphical data having such small variations, as are well known to the skilled person, in comparison with the Figure.

[0025] A solid state form (or polymorph) may be referred to herein as polymorphically pure or as substantially free of any other solid state (or polymorphic) forms. As used herein in this context, the expression “substantially free of any other forms” will be understood to mean that the solid state form contains about 20% (w/w) or less, about 10% (w/w) or less, about 5% (w/w) or less, about 2% (w/w) or less, about 1% (w/w) or less, or about 0% of any other forms of the subject compound as measured, for example, by XRPD.

[0026] Thus, solid state form of Reproxalap salt described herein as substantially free of any other solid state forms would be understood to contain greater than about 80% (w/w), greater than about 90% (w/w), greater than about 95% (w/w), greater than about 98% (w/w), greater than about 99% (w/w), or about 100% of the subject solid state form of Reproxalap salt.

[0027] In some embodiments of the disclosure, the described solid state forms of Reproxalap salt may contain from about 1% to about 20% (w/w), from about 5% to about 20% (w/w), or from about 5% to about 10% (w/w) of one or more other solid state forms of Reproxalap salt.

[0028] As used herein, unless stated otherwise, XRPD peaks reported herein are preferably measured using CuK α radiation, $\lambda=1.5418$ Å, preferably, XRPD peaks reported herein are measured using CuK α radiation, $\lambda=1.5418$ Å, at a temperature of $25\pm3^\circ$ C.

[0029] As used herein, the term “isolated” in reference to the solid state forms of Reproxalap salts of the present disclosure corresponds to solid state form of Reproxalap salt that is physically separated from the reaction mixture in which it is formed.

[0030] A thing, e.g., a reaction mixture, may be characterized herein as being at, or allowed to come to “room temperature”, often abbreviated “RT.” This means that the temperature of the thing is close to, or the same as, that of the space, e.g., the room or fume hood, in which the thing is located. Typically, room temperature is from about 20° C. to about 30° C., or about 22° C. to about 27° C., or about 25° C.

[0031] A process or step may be referred to herein as being carried out “overnight.” This refers to a time interval, e.g., for the process or step, that spans the time during the night, when that process or step may not be actively observed. This time interval is from about 8 to about 20 hours, or about 10 to about 18 hours, typically about 16 hours.

[0032] As used herein, and unless stated otherwise, the term “anhydrous” in relation to a crystalline Reproxalap salt

was added at temperature of about 5° C. to 20° C. and slurry was formed and maintained for 2 hours at same temperature. The obtained solid was filtered under vacuum at temperature of about 20° C. to 30° C., washed with 1 volume to 5 volume of MTBE, and suck dried for period of about 10 minutes to 15 minutes at same temperature, dried at temperature of about 30° C. to 40° C. in vacuum dryer for a period of about 12 hours to 15 hours to obtain Reproxalap methane sulfonate salt.

1. Crystalline Form P1 of Reproxalap Methanesulfonate, characterized by data selected from one or more of the following:

- a. an XRPD pattern having peaks at 8.2, 13.4, 17.7, 18.1, 20.6 and 25.9 degrees 2-theta $\pm$ 0.2 degrees 2-theta;
- b. an XRPD pattern as depicted in FIG. 4; and
- c. combinations of these data.

2. Crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1, characterized by an XRPD pattern having peaks at 8.2, 13.4, 17.7, 18.1, 20.6 and 25.9 degrees 2-theta $\pm$ 0.2 degrees 2-theta, and also having one, two, three, four or five additional peaks selected from 11.0, 15.3, 22.5, 23.2 and 29.3 degrees two theta $\pm$ 0.2 degrees two theta; or characterized by an XRPD pattern having peaks at: 8.2, 11.0, 13.4, 15.3, 17.7, 18.1, 20.6, 22.5, 23.2, 25.9, 29.3 degrees 2-theta $\pm$ 0.2 degrees 2-theta.

3. Crystalline Form P1 of Reproxalap Methanesulfonate crystalline Form R1 according to claim 1, wherein the crystalline form is a hydrate form.

4. Crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1, which is polymorphically pure, or wherein the solid state form contains about 20% or less of any other solid state forms of Reproxalap Methanesulfonate.

5. A pharmaceutical composition comprising Crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1.

6. (canceled)

7. A pharmaceutical formulation comprising Crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1 and at least one pharmaceutically acceptable excipient, wherein the pharmaceutical formulation is an ophthalmic solution, a topical cream, or an ointment.

8. A process for preparing a pharmaceutical formulation according to claim 7, comprising combining the Crystalline Form P1 of Reproxalap Methanesulfonate with at least one pharmaceutically acceptable excipient.

9. A medicament comprising the Crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1.

10. (canceled)

11. (canceled)

12. A method of treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis or Sjögren-Larsson Syndrome comprising administering a therapeutically effective amount of the Crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1, to a subject in need of the treatment.

13. A method of treatment of Sjögren-Larsson Syndrome comprising administering a therapeutically effective amount of the Crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1 to a subject in need of the treatment.

14. (canceled)

15. A process for preparing a Reproxalap salt or a solid state form thereof or Reproxalap base or a solid state form thereof, comprising preparing the crystalline Form P1 of Reproxalap Methanesulfonate according to claim 1, and converting it to another Reproxalap salt or a solid state form thereof, or Reproxalap base or a solid state form thereof.

16. Crystalline Form R1 of Reproxalap borate, characterized by data selected from one or more of the following:

- a. an XRPD pattern having peaks at 13.1, 15.3, 16.0, 16.8 and 23.0 degrees 2-theta $\pm$ 0.2 degrees 2-theta;
- b. an XRPD pattern as depicted in FIG. 1; and
- c. combinations of these data.

17. Crystalline Form R1 of Reproxalap borate according to claim 16, characterized by the XRPD pattern having peaks at 13.1, 15.3, 16.0, 16.8 and 23.0 degrees 2-theta $\pm$ 0.2 degrees 2-theta, and also having one, two, three, four or five additional peaks selected from 11.2, 14.2, 24.2, 24.7 and 25.3 degrees two theta $\pm$ 0.2 degrees two theta; or characterized by an X-ray powder diffraction pattern having peaks at: 11.2, 13.1, 14.2, 15.3, 16.0, 16.8, 23.0, 24.2, 24.7, 25.3 degrees 2-theta $\pm$ 0.2 degrees 2-theta.

18. Crystalline Form R1 of Reproxalap borate according to claim 16, wherein the crystalline form is a hydrate form.

19. Crystalline Form R1 of Reproxalap borate according to claim 16, which is polymorphically pure, or wherein the solid state form contains about 20% or less of any other solid state forms of Reproxalap borate.

20. A pharmaceutical composition comprising Crystalline Form R1 of Reproxalap borate according to claim 16.

21. (canceled)

22. A pharmaceutical formulation comprising Crystalline Form R1 of Reproxalap borate according to claim 16, wherein the pharmaceutical formulation is an ophthalmic solution, a topical cream, or an ointment.

23. A process for preparing the pharmaceutical formulation according to claim 22, comprising combining the Crystalline Form R1 of Reproxalap borate with at least one pharmaceutically acceptable excipient.

24. A medicament comprising the Crystalline Form R1 of Reproxalap borate according to claim 16.

25. (canceled)

26. (canceled)

27. A method of treatment of dry eye disease, allergic conjunctivitis, noninfectious anterior uveitis or Sjögren-Larsson Syndrome comprising administering a therapeutically effective amount of the Crystalline Form R1 of Reproxalap borate according to claim 16 to a subject in need of the treatment.

28. A method of treatment of Sjögren-Larsson Syndrome comprising administering a therapeutically effective amount of the Crystalline Form R1 of Reproxalap borate according to claim 16 to a subject in need of the treatment.

29. (canceled)

30. A process for preparing a Reproxalap salt or a solid state form thereof or Reproxalap base or a solid state form thereof, comprising preparing the Crystalline Form R1 of Reproxalap borate according to claim 16, and converting it to another Reproxalap salt or a solid state form thereof, or Reproxalap base or a solid state form thereof.

* * * * *